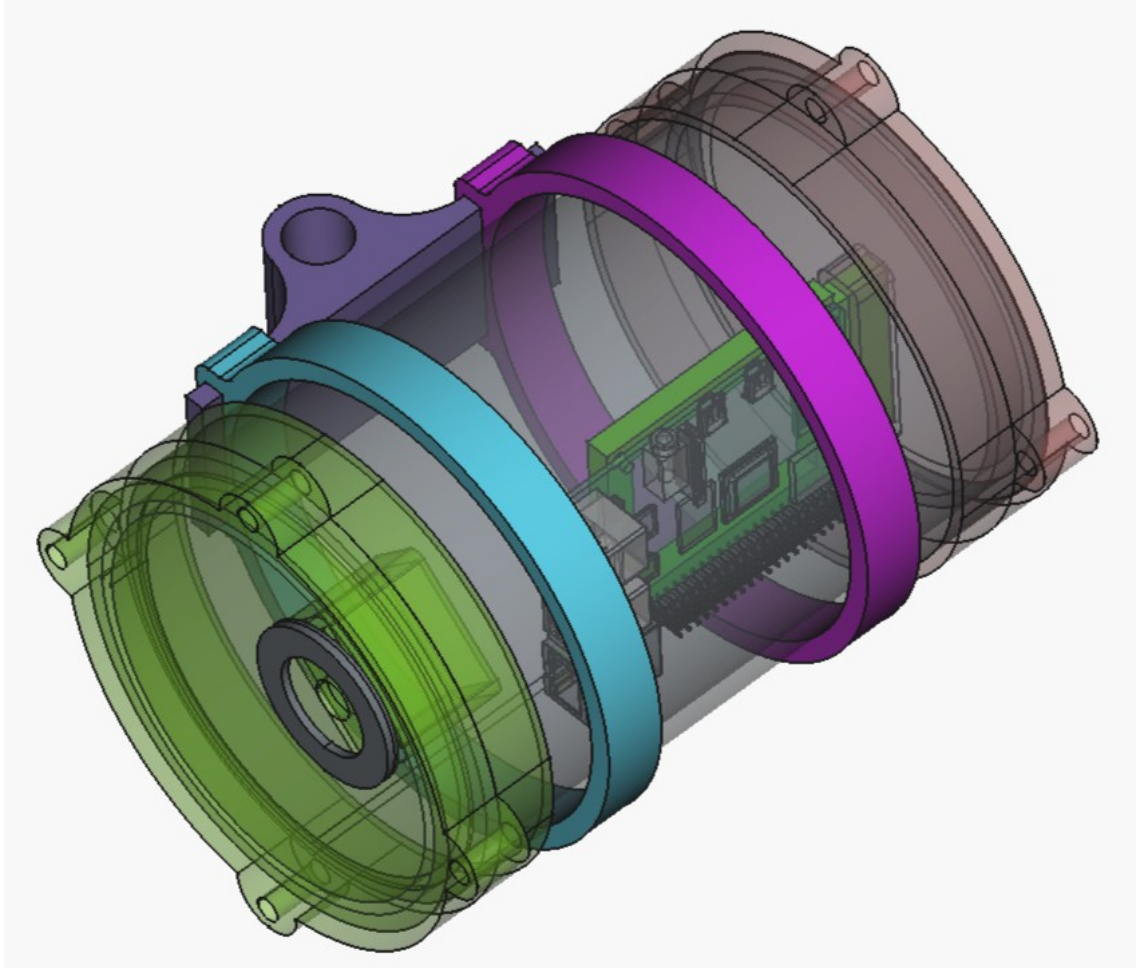




BIRDPipe ASSEMBLY INSTRUCTIONS

AI-CAPABLE SOUND RECORDER BASED ON RASPBERRY PI AND USB AUDIO DEVICES

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BirdPipe is a reproducible field device for autonomous bird sound recording. These instructions describe the required components, enclosure assembly, microphone protection, GNSS and connectivity options, waterproofing considerations, and field-use findings for the first hardware version. This document also includes information about material costs. Note that prices and component availability may vary in the future.

REQUIREMENTS

- Access to a 3D printer or the ability to use commercially available pipe components
- PETG or ABS filament for 3D printing
- USB microphone
- Raspberry Pi 4 Model B (8 GB RAM or more)
- USB cables
- Battery pack, USB charger or 12v USB-C power converter
- 110 mm waste pipe
- Zip ties
- Optional wind protection material, if the microphone does not include a windscreen
- Optional weatherproof acoustic membrane
- Optional mesh netting for insect protection, if needed
- Optional GNSS HAT for Raspberry Pi and antennas
- Optional USB 4G/LTE modem and antennas

COMPONENTS USED IN BIRDPipe DEVICE USED AS EXAMPLE IN THIS DOCUMENT

Component	Used component	Price
Raspberry Pi	Raspberry Pi 4 Model B, 8 GB	€180.00
USB microphone	Dodotronic Ultramic 384K EVO (optional ultrasound)	€280.00
Power source	Wave Ultrapower 30,000 mAh	€39.90
GNSS module	Waveshare GNSS module	€24.99
Mobile connection	Brovi (Huawei) E3372-325 LTE/4G/3G USB	€39.99
Acoustic vent	Gore Acoustic Vent AVP series	€5.00
3D filament	Generic PETG, white	€5.00
110 mm PVC pipe	110 mm waste pipe, 20 cm	€2.00
USB cables	Generic USB cables	€20.00
Total cost per device		€596.88

BIRDPipe SYSTEM OVERVIEW

The example version of BirdPipe device illustrated in this document consists of a commonly used Raspberry Pi 4 Model B (8 GB), a widely used GNSS HAT for time and location synchronization, a USB ultrasonic microphone, open-source Python-based software, and an open enclosure design (Fig. 1). The enclosure can be fabricated using a standard 3D printer, and its open STL models are freely available. The main structural component is a 110 mm plastic waste pipe, a material commonly available worldwide, including as surplus from construction sites. Hardware and software openness and availability are key aspects of the design, ensuring that the system remains transparent, reproducible, and modifiable for research use.

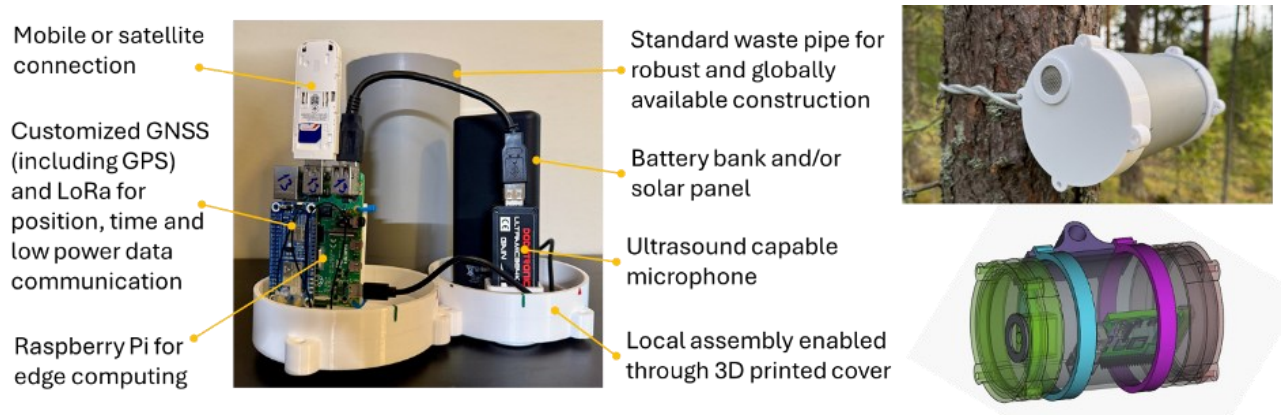


Figure 1. BirdPipe overview.

Any Global Navigation Satellite System (GNSS) module compatible with the Raspberry Pi (Linux) and the `gpsd` GPS service daemon can be used, provided that it supports access to time, location, and optional PPS (Pulse Per Second) signals through the `gpsd` interface. The device can operate without GNSS by using predefined location coordinates, but in that case UTC time must be managed differently, such as through a network connection.

In our study, Dodotronic Ultramic 384K EVO ultrasonic microphones were employed, enabling the detection of both bird and bat vocalizations through an AI model based on TensorFlow Lite. The software includes help and configuration commands that allow users to modify parameters such as the selection of one or more AI models, the microphone input source, the sampling rate, and the recording schedule. It also enables data-handling options such as saving raw WAV audio, applying FLAC (Free Lossless Audio Codec) compression, or running the AI model entirely in RAM. The RAM-only option can be particularly useful in sensitive situations, such as populated urban areas where the recording of human speech is a privacy concern. Recording uncompressed 384 kHz ultrasonic audio directly to a microSD card can be problematic because of the required write speed. Therefore, buffering in RAM and using an M.2 SSD are recommended for reliable long-term use.

BIRDPipe DETECTION MODES

The detection model is configured using BirdPipe's Python-based management software. The initial configuration can be created by running `uv run birdpipe create-config` or edited by running `uv run birdpipe edit-config`. For more details about BirdPipe functionality, use `uv run birdpipe --help`. Full documentation of installing BirdPipe's Python-based software to Raspberry Pi operating system can be found in BirdPipe repository's `readme.md` file.

BirdPipe's configuration commands set default operational parameters such as audio input settings, available sample rates, recording cycles, data storage options, and the AI models to be used. The device can run multiple AI models and store structured GeoJSON data for a recording. The selected microphone can operate in either ultrasonic mode (384 kHz) or standard mode (48 kHz), and the duration of each recording is user-defined. Audio data may be stored temporarily in memory (RAM) or permanently on disk in WAV or FLAC format. The prediction threshold controls the model's confidence level, and filename options define the naming structure of recorded audio files. The configuration also includes options for testing existing audio files with AI models and specifying predefined coordinates.

If the system is intended to record, for example, for one minute every 15 minutes over a 24-hour period, a systemd service and timer must be created to execute `uv run birdpipe record`. For more detailed instructions on setting up systemd services and timers, see the systemd documentation. After configuring the device, test it and verify that data are accumulating in the newly created folders under the device's home directory. Use `uv run birdpipe record` or the configured systemd service to perform a test recording and analysis. For orchestration and monitoring, we used in our field experiments Tailscale, a mesh VPN, for secure remote access over SSH.

SHELL STRUCTURE AND MOUNTING RASPBERRY PI

Shell consists of the following four components:

- 110 mm waste pipe
- `printable_parts_and_instructions/BirdPipe-110mmEndForDodotronic.stl`
- `printable_parts_and_instructions/BirdPipe-110mmEndForRaspHolder.stl`
- `printable_parts_and_instructions/BirdPipe-RaspberryPiHolder.stl`

The cost of a 110 mm PVC waste pipe in Finland is approximately 2,00 € for a 20 cm section. Only 20 cm of pipe is required for a single BirdPipe unit, making the housing material cost very low. The pipe can be cut using a handsaw, jigsaw, or dedicated pipe-cutting tool. When cutting or handling the pipe, always use appropriate protective equipment, including safety glasses and protective gloves. Remove and round any sharp edges with sandpaper after cutting to prevent injury and ensure proper fit of the assembled components.

The required parts can be printed from the STL files listed above using a 3D printer capable of printing PETG or ABS. We recommend using white filament to reduce heat absorption from direct sunlight. For the BirdPipe prototype assembly described in this document, all parts were printed using a Bambu Lab X1 Carbon 3D printer.

Alternatively, standard off-the-shelf pipe end caps and fittings can be used in place of the 3D-printed parts. This eliminates the need for a 3D printer and allows the enclosure to be assembled using commercially available plumbing components. The selection of suitable fittings depends on the pipe type and dimensions

available locally. Before purchasing materials, verify the pipe dimensions and end-cap compatibility to ensure a secure fit and adequate weather protection.

The end cap and holder components are designed for press-fit assembly, allowing easy installation and removal. Under normal circumstances, no glue or adhesive is required.



Figure 2. Shell components.

Mount the Raspberry Pi on the BirdPipe-RaspberryPiHolder using small zip ties passed through the mounting holes of the Raspberry Pi circuit board. In most cases, fastening two opposite corners provides adequate support and keeps the board securely positioned inside the enclosure. Ensure that the zip ties do not interfere with connectors, cables, or electronic components.

During assembly, ensure that the USB ports face upwards, as shown in the relevant assembly image. Insert the microSD card before tightening the zip ties.

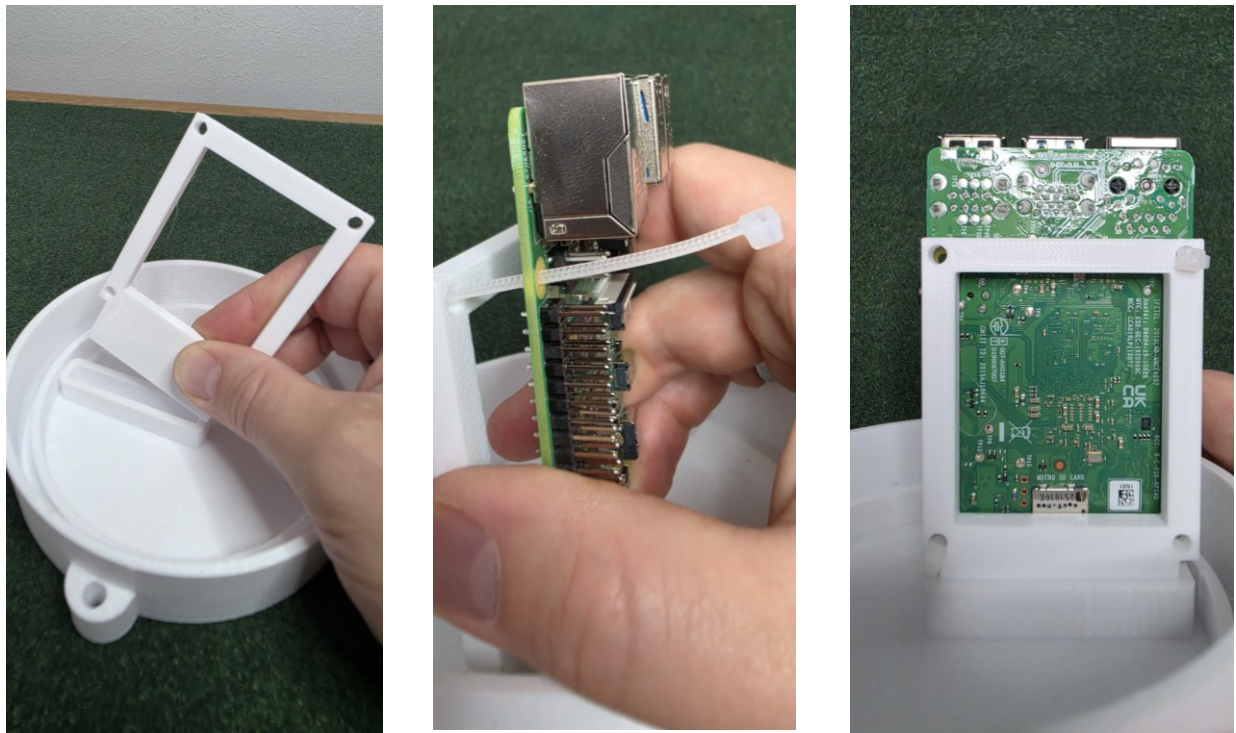


Figure 3. Raspberry Pi mount (left-hand panel). Zip ties used for attaching the Raspberry Pi (middle panel). Completed assembly (right-hand panel).

MICROPHONE AND AUDIO INPUT OPTIONS

In our device, we used the Dodotronic Ultramic 384K EVO, an ultrasonic microphone capable of recording frequencies beyond the range of human hearing. The microphone provides excellent audio quality but is relatively expensive, costing approximately 280,00 €. However, BirdPipe is based on Linux and is not limited to this microphone model. Most USB microphones compatible with Raspberry Pi and Raspberry Pi OS can also be used for audio recording, allowing users to choose a more affordable option depending on their requirements.

Possible audio input alternatives include a Sound Blaster USB audio interface with an analogue microphone, an inexpensive USB microphone, or the Dodotronic Ultramic 384K EVO.

The structure `printable_parts_and_instructions/BirdPipe-110mmEndForDodotronic.stl` holds the Dodotronic microphone. With suitable tools and assembly skills, the structure can be adapted for another microphone variant to reduce costs considerably.



Figure 4. Dodotronic connected to the Raspberry Pi (left-hand panel). Alternative microphone model connected (middle panel). USB sound card and 3.5 mm jack analogue microphone (right-hand panel).

Another cable is used for USB power. In our prototype, we used a 30,000 mAh power bank. Alternatively, the device can be powered from a wall outlet using a USB power adapter or from a solar-powered system using an appropriate USB or 12 V power converter. It is important to understand the safety considerations associated with batteries, solar panels, and electrical installations. Always follow the manufacturer's instructions and applicable local regulations. If you are unsure about any aspect of the installation, consult a qualified local specialist.

MICROPHONE PROTECTION

The open microphone hole is a vulnerable part of the enclosure. Prevent water, dust, insects, and other debris from entering the device through the microphone opening. The BirdPipe enclosure is not fully waterproof, and additional protection may be required depending on the installation and research environment.

A Gore Acoustic Vent from the AVP series, approximately 1 cm in diameter, can protect the opening from rain and water splashes. In the prototype, the corners of the membrane were secured with a small amount of general-purpose cyanoacrylate adhesive. Confirm the membrane manufacturer's installation requirements before applying adhesive.

A small amount of Superlon-type foam can be used as a windscreen for the microphone.

To help prevent insects from entering the device, a piece of fine mesh can be installed over the opening together with MicrophoneProtector.stl. Alternatively, one half of a tea strainer can provide a convenient and inexpensive insect screen.

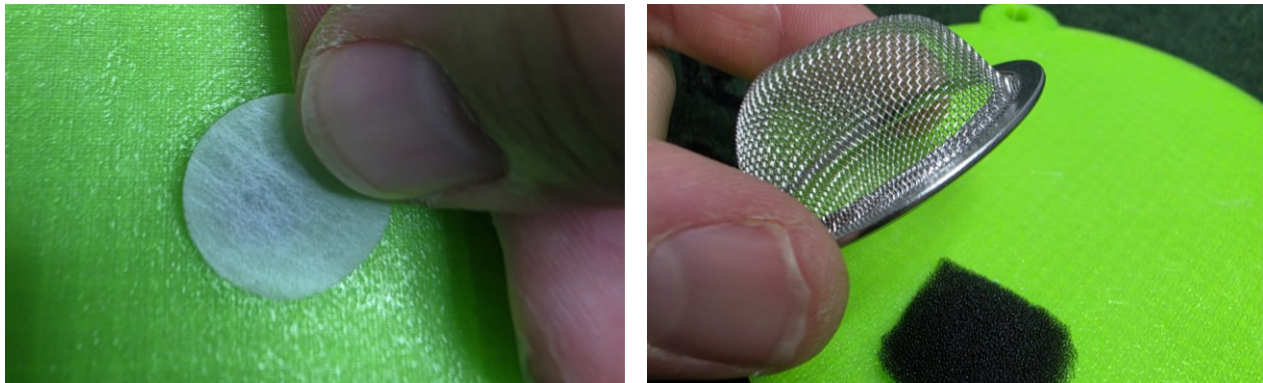


Figure 5. Gore Acoustic Vent AVP series, approximately 1 cm in diameter (left-hand panel). One half of a tea strainer provides a convenient and inexpensive insect screen (right-hand panel).

Sensitive microphones can be affected by electromagnetic interference and acoustic resonance inside the enclosure. To reduce unwanted noise, isolate the microphone from vibrating surfaces. The Dodotronic microphone mount provides a tight fit that minimizes acoustic resonance. If an optional LTE modem is used, place it outside the enclosure and as far from the microphone as practical to reduce potential radio-frequency interference.

GNSS – SATELLITE BASED POSITIONING

Any Global Navigation Satellite System (GNSS) module compatible with the Raspberry Pi and the gpsd GPS service daemon can be used, provided that it supports access to time, location, and PPS (Pulse Per Second) signals through the gpsd interface. The device can operate without GNSS by using predefined location coordinates, but in that case UTC time must be managed differently, such as through a network connection.

GNSS functionality can be added using a GNSS receiver. In addition to providing accurate location data, GNSS supplies a precise time reference even where no internet connection is available. Several GNSS modules are available for the Raspberry Pi platform and can be installed as HAT expansion boards. In this project, a Waveshare module is used. Integrating GNSS improves the temporal accuracy of audio recordings and makes deployed devices easier to locate in the field.

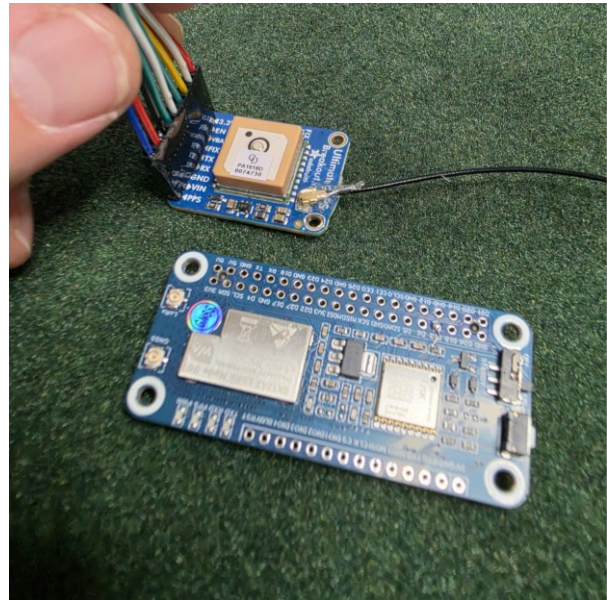


Figure 6. Waveshare SX1262 LoRa and GNSS board assembled on the Raspberry Pi using GPIO pins (left-hand panel). Waveshare SX1262 and a compact GNSS alternative with an integrated ceramic antenna (right-hand panel).

External antennas can be mounted at a suitable distance from the device using a structure similar to the one shown in the assembly images. Cable pass-through holes for antenna connections should be drilled in the rear end cap. Positioning antennas outside the enclosure and away from sensitive electronics can improve signal quality and reduce interference.



Figure 7. Two examples of antenna mount.

Files required for antenna assembly:

- external-antenna-structure-LTE-holder.stl
- external-antenna-structure-Plate.stl
- external-antenna-structure-Tree-Attachment.stl

POWER MANAGEMENT

The power consumption of BirdPipe's Raspberry Pi based infrastructure is a critical design consideration. Unlike passive acoustic monitors, BirdPipe utilizes a Raspberry Pi to perform edge computing with ultrasound capabilities, which is inherently power-intensive. In standard configurations using a 30,000 mAh power bank, the device is designed for short-term deployments, with a typical runtime of approximately +24-48 hours depending on the environmental variables.

A significant driver of this energy consumption is the LTE connectivity and edge computing, especially in areas with limited cellular network signals. The power required for wireless data transmission of metadata and audio classifications impacts the battery life substantially. We acknowledge the trade-off between the ability to perform real-time biodiversity monitoring and the resulting energy footprint.

For deployments requiring longer autonomous operation, we recommend replacing the portable power bank with a more robust power system. A 12V AGM battery combined with appropriate solar charge controllers and a solar panel provides a sustainable energy solution for extended field research needs. This configuration mitigates the limitations of the Raspberry Pi's power draw and ensures continuous operation in remote areas.

It should be noted that if the research application only requires audio recording without the need for edge computing or wireless data transform, BirdPipe may be over-engineered for the task. In such cases, specialized low-power recording alternatives are likely more efficient options.

WATERPROOFING CONSIDERATIONS

The device is not completely waterproof. The following measures are recommended to improve its resistance to water:

- Seal all external cable entries.
- Silicone sealant can be applied to the groove of the end caps; however, the pipe must be cut accurately to ensure a proper seal.
- Additional waterproof tape can be applied over the enclosure joints for extra protection.
- If there is no need to open the front end cap during operation, it can be permanently glued in place. In that case, only the rear end cap should remain removable for maintenance and access.

DATA CONNECTIVITY

If a cellular, satellite, LAN or WLAN connection has been added to the BirdPipe device, observation data, such as GeoJSON files and/or audio files, can optionally be transferred using rclone to a cloud service, an organization's S3 storage, or an FTP server. From there, the data can be retrieved almost in real time for analysis with the BirdPipe R-package. This makes it possible to monitor observations and the state of data synchronization without physically accessing the device. When cloud services or other remote storage solutions are used, data protection and information security must be carefully considered, especially if audio recordings or location data are transferred outside the device.

MOUNTING AND SECURITY

The device can be mounted using the holes integrated into the end caps and a rubber-coated steel cable, for example to a tree, pole, or similar structure. The cable can be locked, which helps prevent unauthorized access to the device and reduces the risk of component theft.



Figure 8. BirdPipe mounted in the field.